

Examiner reports

Q1.

The printed answer helped candidates to do well in part (a). A few candidates did not show enough working to justify the allocation of all three marks. There were some candidates who worked with equations for the vertical component of the motion, rather than the horizontal component.

Part (b) was again well answered. A few candidates made arithmetic errors and some found the maximum height instead of the height of the crossbar.

Candidates found part (c) very difficult. Many of them considered the position of the ball rather than its velocity. Some considered only one component of the velocity. However there were a few candidates who did produce a good solution to this part of the question. Most of the candidates stated appropriate assumptions, although some candidates produced inappropriate assumptions.

Q2.

The candidates were more successful with this question than had been expected.

Part (a) was quite well answered. A reasonable number of candidates formed the two equations of motion correctly and then went on to obtain the required value. Some candidates did not resolve the weight of the particle on the slope. A small number of candidates did not form an equation of motion for each particle.

Those candidates who realised that the acceleration was zero did well in part (b). There were quite a few candidates who tried to use the acceleration from part (a) when tackling part (b).

Q3.

Part (a) was done well by many candidates, who may have been helped by the printed answer. There were also some candidates who made little progress and did not seem to be able to see how to form an equation to find the time. Part (b) was often done well, but was completed by fewer candidates than part (a). Some candidates did not seem to be able to form a correct expression for the height. A few candidates used

$$h = 10 \sin 60^\circ t + \frac{1}{2} g t^2$$
 instead of

$$h = 10 \sin 60^\circ t - \frac{1}{2} g t^2$$
 . Part (c) proved to be very challenging.

Many candidates made very little progress with this part of the question. Some found the vertical component of the velocity when the ball hit the wall. Only a small proportion of the candidates could incorporate the horizontal component and find the correct speed.

Q4.

The force diagram asked for in this question was drawn fairly well, but the usual errors of forces in the direction of motion and missing forces were present on some scripts. Parts (b) and (c) were done fairly well and it appeared that many 'candidates' had been helped by the printed answer in part (b). There were some candidates who did not realise that they should consider the tension in the rope when finding the normal reaction. Part (d)

was found to be very much more demanding, with relatively few correct responses. Candidate typically omitted one term from their equation or failed to solve correctly for T .

Q5.

There were several good solutions to part (a), but some candidates ignored the instruction to use two equations of motion. Some candidates assumed that A would move up the slope and then ran into difficulties with the negative signs that were involved. Many candidates found part (b) difficult. It was very common for them to try to use the acceleration from part (a) instead of setting the acceleration equal to zero. Interestingly, some candidates who made little or no progress with part (a) did complete part (b) successfully.

Q6.

In this question, there were problems with the force diagram and resolving correctly. Many candidates were able to find the reaction force correctly, but there were more problems in part (d).

Q7.

The candidates who knew how to tackle projectile questions with confidence did well with this question. Part (a) did not present much difficulty. However, in part (b), including the correct signs and solving the quadratic equation correctly were more challenging tasks. There were some very poor attempts at solving the quadratic equation, where candidates did not realise that the quadratic equation formula was required.

Q8.

This question was done quite well on the whole. The major source of errors was resolving incorrectly. The fact that the friction force was given did help some candidates to work towards the correct answer.

Q9.

A lot of the candidates found this question very difficult. There were a number of good clear force diagrams, but there were many that included a friction force. Some also introduced a force acting down the slope. A few candidates did not make the directions of the forces clear. Many candidates were unable to resolve horizontally in part (b). The most common error was to resolve the 8 N force instead of the reaction force, obtaining equations such as $R = 8 \cos 30^\circ$. In part (c), many candidates produced results such as $R = mg \cos \theta$, by resolving perpendicular to the slope and ignoring the 8 N force. A few simply stated that $R = mg$.

Q10.

There were a fair number of good responses to part (a). The candidates who did not produce a correct solution generally either missed out the difference between the initial and final heights of the ball, or made sign errors in the setting up of their equations. A few candidates made arithmetic errors. Part (b) was also done well, with some candidates being helped by the printed answer in part (a). A few candidates used incorrect times from part (a) when trying to find the range. There were only a few correct solutions to part (c). An incomplete solution, which only found the vertical component of the velocity, was quite

common among those who made some progress with the solution.

Q11.

In part (a) of this question, some candidates used an incorrect angle when trying to resolve. Some others simply assumed that the normal reaction force was equal to the weight. In part (b), those who had had difficulties resolving in part (a) had further difficulties here. However there were a reasonable number of candidates who were able to find the correct coefficient of friction. A large number of candidates did not include an inequality in their working. In part (c), a good number of candidates calculated the magnitude of the friction force correctly, but some then went on to assume that this was the only force acting.

Q12.

Part (a) was answered quite well, with many candidates making good use of the printed answer.

Part (b) caused more problems. Some candidates only found the vertical component of the velocity and did not bring the horizontal component into their working. As a result there were relatively few correct answers for the speed, even though quite a few candidates found the vertical component correctly. There were also quite a number of candidates you did not really know how to use the height to find a time or a velocity.

Q13.

Drawing the force diagram in part (a) did cause difficulties for the candidates with several wanting to include a force directed down the slope. Part (b) was done correctly by almost all of the candidates. In part (c), several candidates did not form a three term equation, either omitting the component of the weight down the slope or the friction force. Most candidates produced a good attempt at part (d).

Q14.

Part (a) was either done very well or very badly. Some candidates produced simple, correct arguments. Two common incorrect approaches were to take $v = V \sin \alpha$ instead of $u = V \sin \alpha$, or to omit the negative sign from the acceleration when using the equation as $v^2 = u^2 + 2as$. The other approach that caused difficulty was when candidates tried to find

the time for the maximum height by working with the expression $y = V \sin \alpha t - \frac{1}{2}gt^2$. This led to a variety of problems. The candidates who used the formula from part (a) usually had no difficulty finding V . A number of candidates did not use this result and often ended up with the vertical component of the initial velocity instead of the required speed. Part (c) caused problems for some candidates, the biggest problem being finding the time of flight correctly. Some used an equation that would have given the time for the maximum height by considering $y = 6$.

Q15.

Part (a) was done very well and many candidates were able to obtain the acceleration given in the question from correct working. Part (b) proved to be more problematic. The equation of motion was often set up incorrectly, the negative acceleration being a

particular problem for some. In part (b), a number of candidates subtracted the 300 N from their previous answer rather than adding it on. Some candidates did not realise that they could simply modify their answer to part (b) and did a lot more work than was needed. There were some good answers to part (d), often expressed well. There were also a number of candidates who gave spurious answers.

Q16.

There were many good solutions to this question. There were a number of candidates who included additional forces on their force diagrams. Candidates should be encouraged to only show the forces acting on force diagrams. Including velocities or components as well as the forces often confuses diagrams and can cause marks to be lost. Part (b) was done very well. In part (c), the most common error was only to include one force, typically the friction force, and not to include the component of the weight that acts down the slope. Two other less common errors were to use $80ga$ instead of $80a$ and to resolve the weight incorrectly or not at all.

Q17.

There were many good responses to this question, with many candidates scoring full marks. Some candidates resolved horizontally and vertically, while others used a vector triangle approach combined with the sine rule. The most common error was to resolve incorrectly or use the wrong angles.

Q18.

While there were some good proofs of the formula for the range, there were also a number of confused approaches. Several of these were based on the formula as $v^2 = u^2 + 2as$. However, those who found the time of flight first often made quite good progress. Some candidates misquoted the double angle formula. Another area of difficulty was in the solution of the quadratic equation to find the time of flight.

In part (b), almost all of the candidates gave the answer 45° , but very few were able to give a convincing argument.

Part (c) was generally done very well. Some candidates tried to do this without using the results from parts (a) and (b). Many of these candidates were unsuccessful. A few candidates did not give their answer for the range to the nearest metre as instructed in the question.

Q19.

Part (a) was generally done well. Some candidates made arithmetic errors and a small number misquoted the constant acceleration equation. The force diagrams were drawn very well. Most of the candidates found it easy to calculate the magnitude of the friction force, although a small number took the reaction force as being equal to the weight. Many candidates had difficulty finding the tension in the rope.

Q20.

There were a number of good responses to this question. Some parts caused the candidates a little more difficulty than the earlier questions. In part (a), some candidates included additional forces. Parts (b) and (c) were generally well done, the only significant

problem was that some candidates used the wrong angle when finding the components of the weight and so ended up with their answers to parts (b) and (c) interchanged. This caused subsequent problems in part (d). However, most candidates made a good attempt at part (d). Part (e) was more challenging. Many candidates felt that the 14 kg would just slide down the slope. Some argued that the friction and weight would be proportional to the mass and so it would stay still. Quite a number repeated all the earlier calculations to obtain the same coefficient of friction and often concluded that the block would remain at rest.

Q21.

There were a number of good solutions to part (a). Some candidates did make sign errors when setting up the equation for the vertical motion. Some candidates broke the motion down into stages, but those that did often made small errors. Part (b) was done very well, with some candidates using the time given in part (a) to enable them to find the range. Part (c) proved to be very demanding. Some candidates calculated the vertical component of the velocity when the car hit the sea, but did not bring the horizontal component into their solutions. There were only very few complete solutions.



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